



DEPARTMENT OF COMPUTER SCIENCE
CYBER SECURITY

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DIGITAL FORENSICS LAB

STEGNOGRAPHY

Task 1: Verify Steganography Tools

STEGHIDE

INSTALLATION

```
(kali@kali)-[~]
$ steghide
Command 'steghide' not found, but can be installed with:
sudo apt install steghide
Do you want to install it? (N/y)y
sudo apt install steghide
[sudo] password for kali:
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Installing:
  steghide

Installing dependencies:
  libmccrypt4 libmhash2

Suggested packages:
  libmccrypt-dev mcrpyrt

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 1879
  Download size: 309 kB
  Space needed: 916 kB / 4,108 MB available

Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libmccrypt4 amd64 2.5.8-8+b1 [72.1 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libmhash2 amd64 0.9.9-11 [93.2 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 steghide amd64 0.5.1-15 [144 kB]
Fetched 309 kB in 1s (342 kB/s)
Selecting previously unselected package libmccrypt4:amd64.
(Reading database ... 422523 files and directories currently installed.)
Preparing to unpack .../libmccrypt4_2.5.8-8+b1_amd64.deb ...
Unpacking libmccrypt4:amd64 (2.5.8-8+b1) ...
Selecting previously unselected package libmhash2:amd64.
Preparing to unpack .../libmhash2_0.9.9-11_amd64.deb ...
Unpacking libmhash2:amd64 (0.9.9-11) ...
Selecting previously unselected package steghide.
Preparing to unpack .../steghide_0.5.1-15_amd64.deb ...
Unpacking steghide (0.5.1-15) ...
Setting up libmhash2:amd64 (0.9.9-11) ...
Setting up libmccrypt4:amd64 (2.5.8-8+b1) ...
Setting up steghide (0.5.1-15) ...
Processing triggers for libc-bin (2.42-13) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

(kali@kali)-[~]
$
```

Version

```
(kali㉿kali)-[~]  
$ steghide version  
steghide version 0.5.1
```

BINWALK

INSTALLATION

```
(kali㉿kali)-[~]  
$ sudo apt install steghide binwalk exiftool -y  
Note, selecting 'libimage-exiftool-perl' instead of 'exiftool'  
steghide is already the newest version (0.5.1-15).  
binwalk is already the newest version (2.4.3+dfsg1-2).  
binwalk set to manually installed.  
The following package was automatically installed and is no longer required:  
  libcrypt-dev  
Use 'sudo apt autoremove' to remove it.  
  
Upgrading:  
  libimage-exiftool-perl  
  
Summary:  
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 1878  
  Download size: 5,960 kB  
  Space needed: 317 kB / 4,107 MB available  
  
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libimage-exiftool-perl all 13.50+dfsg-1 [5,960 kB]  
Fetched 5,960 kB in 5s (1,253 kB/s)  
(Reading database ... 422554 files and directories currently installed.)  
Preparing to unpack .../libimage-exiftool-perl_13.50+dfsg-1_all.deb ...  
Unpacking libimage-exiftool-perl (13.50+dfsg-1) over (13.36+dfsg-1) ...  
Setting up libimage-exiftool-perl (13.50+dfsg-1) ...  
Processing triggers for doc-base (0.11.2) ...  
Processing 42 changed doc-base files ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for kali-menu (2025.4.3) ...
```

Version

```
(kali㉿kali)-[~]  
$ binwalk  
  
Binwalk v2.4.3  
Original author: Craig Heffner, ReFirmLabs  
https://github.com/OSPG/binwalk
```

EXIFTOOL

INSTALLATION

```
(kali㉿kali)-[~/Desktop]
$ sudo apt update
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1968 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali㉿kali)-[~/Desktop]
$
```

```
(kali㉿kali)-[~/Desktop]
$ sudo apt install exiftool -y
Note, selecting 'libimage-exiftool-perl' instead of 'exiftool'
libimage-exiftool-perl is already the newest version (13.50+dfsg-1).
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1968

(kali㉿kali)-[~/Desktop]
$
```

Version

```
(kali㉿kali)-[~/Desktop]
$ exiftool -ver
13.50
```

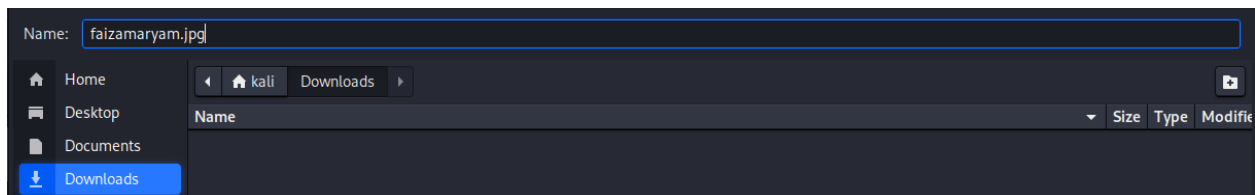
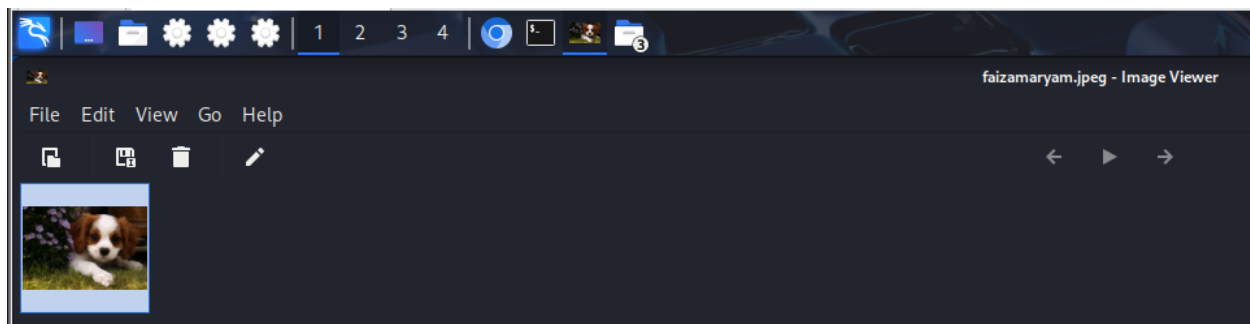
Task 2: Hide a Text File inside an Image (Steghide)

Step 1: Create a Secret File

```
(kali㉿kali)-[~]
$ echo "This is the secret file" > secret.txt
```

Step 2: Embed the File

To embed the file first download jpeg file



```
(kali@kali)-[~]
$ steghide embed -cf faizamaryam.jpg -ef secret.txt -sf faizamaryam_stego.jpg
Enter passphrase:
Re-Enter passphrase:
embedding "secret.txt" in "faizamaryam.jpg"... done
writing stego file "faizamaryam_stego.jpg"... done
```

Paraphrase: 9090

Step 3: Verify

```
(kali@kali)-[~/Desktop]
$ ls
faizamaryam.jpeg  output.jpeg

(kali@kali)-[~/Desktop]
$
```

Task 3: Extract Hidden Data

```
(kali@kali)-[~/Desktop]
$ steghide extract -sf output.jpeg
Enter passphrase:
wrote extracted data to "secret.txt".
```

```

(kali㉿kali)-[~/Desktop]
└─$ ls
faizamaryam.jpeg  output.jpeg  secret.txt

(kali㉿kali)-[~/Desktop]
└─$ cat secret.txt
SECRET FILE

```

Task 4: Hide Data WITHOUT Password.

```

(kali㉿kali)-[~/Desktop]
└─$ steghide embed -cf ~/Desktop/faizamaryam.jpeg -ef secret.txt -sf output_no_pass.jpeg -p ""
embedding "secret.txt" in "/home/kali/Desktop/faizamaryam.jpeg" ... done
writing stego file "output_no_pass.jpeg" ... done

(kali㉿kali)-[~/Desktop]
└─$

```

Task 5: Check Embedded Data Information

```

(kali㉿kali)-[~/Desktop]
└─$ steghide info output.jpeg
"output.jpeg":
  format: jpeg
  capacity: 593.0 Byte
Try to get information about embedded data ? (y/n) y
Enter passphrase:
  embedded file "secret.txt":
    size: 12.0 Byte
    encrypted: rijndael-128, cbc
    compressed: yes

(kali㉿kali)-[~/Desktop]
└─$

```

Task 6: Detect Hidden Data Using Binwalk

```

compressed: yes

(kali㉿kali)-[~/Desktop]
└─$ binwalk output.jpeg

```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, JFIF standard 1.01


```
(kali@kali)-[~/Desktop]
$ binwalk -e output.jpeg

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
WARNING: One or more files failed to extract: either no utility was found or it's unimplemented

(kali@kali)-[~/Desktop]
$
```

ExifTool

```
(kali@kali)-[~/Desktop]
$ exiftool output.jpeg
ExifTool Version Number      : 13.50
File Name                    : output.jpeg
Directory                   : .
File Size                    : 11 kB
File Modification Date/Time  : 2026:05:04 05:20:36-04:00
File Access Date/Time       : 2026:05:04 05:20:36-04:00
File Inode Change Date/Time  : 2026:05:04 05:20:36-04:00
File Permissions             : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : None
X Resolution                 : 1
Y Resolution                 : 1
Image Width                  : 275
Image Height                 : 183
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 275x183
Megapixels                   : 0.050
```

The image appears normal in structure, but a few signs are suspicious. The file size is very small (11 KB), metadata fields like author or comments are missing, and the modification time is recent. These factors may indicate that the image was recently altered and could contain hidden data.

Task 8: Compare Original vs Stego Image

```
(kali㉿kali)-[~/Desktop]
$ ls -lh ~/Desktop/faizamaryam.jpeg output.jpeg
-rw-rw-r-- 1 kali kali 9.8K May  4 05:11 /home/kali/Desktop/faizamaryam.jpeg
-rw-rw-r-- 1 kali kali 11K May  4 05:20 output.jpeg

(kali㉿kali)-[~/Desktop]
$
```

Task 9: Audio Steganography

Installing SOX

```
(kali㉿kali)-[~/Desktop]
$ sudo apt install sox -y
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Installing:
  sox

Installing dependencies:
  libid3tag0 libmad0 libsox-ng3

Suggested packages:
  ffmpeg libsox-fmt-alsa | libsox-fmt-oss

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 1968
  Download size: 603 kB
  Space needed: 1,476 kB / 3,945 MB available
```

```
Suggested packages:
  ffmpeg libsox-fmt-alsa | libsox-fmt-oss

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 1968
  Download size: 603 kB
  Space needed: 1,476 kB / 3,945 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 libid3tag0 amd64 0.16.3-4 [35.4 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libsox-ng3 amd64 14.7.1.1+ds1-1 [355 kB]
Get:4 http://http.kali.org/kali kali-rolling/main amd64 sox amd64 14.7.1.1+ds1-1 [147 kB]
Get:2 http://mirror.ourhost.az/kali kali-rolling/main amd64 libmad0 amd64 0.16.4-3 [65.7 kB]
Fetched 603 kB in 6s (100 kB/s)
Selecting previously unselected package libid3tag0:amd64.
(Reading database ... 422567 files and directories currently installed.)
Preparing to unpack .../libid3tag0_0.16.3-4_amd64.deb ...
Unpacking libid3tag0:amd64 (0.16.3-4) ...
Selecting previously unselected package libmad0:amd64.
Preparing to unpack .../libmad0_0.16.4-3_amd64.deb ...
Unpacking libmad0:amd64 (0.16.4-3) ...
Selecting previously unselected package libsox-ng3:amd64.
Preparing to unpack .../libsox-ng3_14.7.1.1+ds1-1_amd64.deb ...
Unpacking libsox-ng3:amd64 (14.7.1.1+ds1-1) ...
Selecting previously unselected package sox.
Preparing to unpack .../sox_14.7.1.1+ds1-1_amd64.deb ...
Unpacking sox (14.7.1.1+ds1-1) ...
Setting up libid3tag0:amd64 (0.16.3-4) ...
Setting up libmad0:amd64 (0.16.4-3) ...
Setting up libsox-ng3:amd64 (14.7.1.1+ds1-1) ...
Setting up sox (14.7.1.1+ds1-1) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for libc-bin (2.42-13) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for mailcap (3.75) ...
```


Downloading Audio

```
(kali㉿kali)-[~/Desktop]
$ sox -n -r 44100 -c 1 -e signed-integer -b 16 faiza_audio.wav trim 0.0 5.0

(kali㉿kali)-[~/Desktop]
$
```



Embed

```
(kali㉿kali)-[~/Desktop]
$ steghide embed -cf faiza_audio.wav -ef secret.txt -sf audio_stego.wav
Enter passphrase:
Re-Enter passphrase:
embedding "secret.txt" in "faiza_audio.wav" ... done
writing stego file "audio_stego.wav" ... done
```

Extract

```
(kali㉿kali)-[~/Desktop]
$ steghide extract -sf audio_stego.wav
Enter passphrase:
the file "secret.txt" does already exist. overwrite ? (y/n) y
wrote extracted data to "secret.txt".
```

Task 10: Encrypt + Steganography (Advanced)

Encryption

```
(kali㉿kali)-[~/Desktop]
$ openssl enc -aes-256-cbc -pbkdf2 -salt -in secret.txt -out secret.enc
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
```

Embed encrypted file into image

```
(kali㉿kali)-[~/Desktop]
$ steghide embed -cf faizamaryam.jpeg -ef secret.enc -sf stego.jpeg
Enter passphrase:
Re-Enter passphrase:
embedding "secret.enc" in "faizamaryam.jpeg" ... done
writing stego file "stego.jpeg" ... done
```

Extract hidden encrypted file

```
(kali㉿kali)-[~/Desktop]
$ steghide extract -sf stego.jpeg
Enter passphrase:
the file "secret.enc" does already exist. overwrite ? (y/n) y
wrote extracted data to "secret.enc".
```

```
(kali㉿kali)-[~/Desktop]
$
```

Decrypt

```
(kali㉿kali)-[~/Desktop]
$ openssl enc -d -aes-256-cbc -pbkdf2 -in secret.enc -out secret.txt
enter AES-256-CBC decryption password:
```

Task 11: Research based Task

1. Difference between Steganography and Cryptography

Cryptography and steganography are both techniques used to protect information, but they work in different ways:

- **Cryptography** hides the content of a message by converting it into an unreadable format. Even if someone sees the data, they cannot understand it without a key.
- **Steganography** hides the existence of the message itself by embedding it inside another file or medium (image, audio, video, etc.).
 - Example: hiding a secret message inside a picture so it looks normal.

Key difference:

- Cryptography = hides meaning
- Steganography = hides existence

2. Why is Passphrase Important?

A passphrase is a longer and more secure version of a password, often made of multiple words.

Its importance includes:

- **Stronger security:** Harder to guess or crack than short passwords
- **Resistance to brute-force attacks:** Longer combinations increase complexity
- **Better memorability:** Easier to remember than random characters
- **Used in encryption systems:** Protects private keys and encrypted data

3. How Can Steganography Be Detected?

Detecting steganography is known as steganalysis.

Common methods include:

- **Statistical analysis:** Detects unusual patterns in files (especially images)
- **File size comparison:** Hidden data may increase file size unexpectedly
- **Metadata inspection:** Checking hidden or altered metadata in files
- **Visual inspection:** Slight distortions in images or audio signals
- **Special tools:** Software that scans for hidden payloads (e.g., binwalk, steghide detection tools)

Advanced steganography can be very difficult to detect without specialized tools.

4. Role of Binwalk in Forensics

Binwalk is a powerful tool used in digital forensics and reverse engineering.

Its main roles include:

- **Analyzing firmware images:** Extracting embedded files from firmware
- **Detecting hidden data:** Finds compressed or encrypted files inside other files
- **File carving:** Extracts hidden components from binary data
- **Security analysis:** Helps identify malware or suspicious embedded content

Example use:

A firmware file may contain hidden ZIP or image files that binwalk can extract.

5. Is Steganography Illegal?

Steganography itself is **not illegal** in most countries, including Pakistan and others.

However, legality depends on how it is used:

- **Legal uses:**
 - Digital watermarking
 - Copyright protection
 - Secure communication
 - Privacy protection
- **Illegal uses:**
 - Hiding malware
 - Concealing criminal communication
 - Cyber espionage or data theft